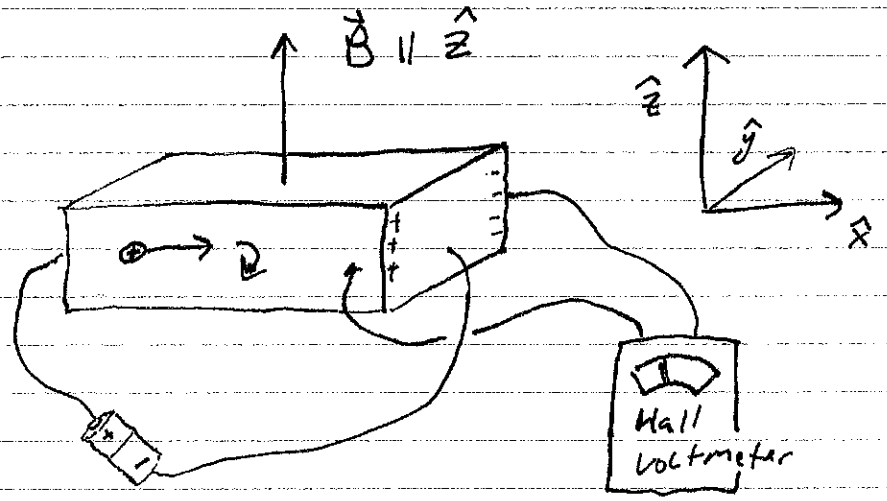
Drüde model description of Hall effect (discussion 4/21/16)



DC voltage drives current through a p-type semiconductor

$\vec{E}_0$ due to battery $= E_0 \hat{x}$

Drüde model $\rightarrow$ $m \frac{dV}{dt} = q E + q V \times B - m\gamma V$

or $\gamma = \frac{1}{\tau}$

at equilibrium $\frac{dV}{dt} = 0$

useful definitions

$$J = e^+ n V = \sigma (E + V \times B)$$

holes

$$\Omega_c = \frac{qB}{m}$$

initially lets not consider finite size of conductor (i.e no Hall voltage)

so using $\frac{dV}{dt} = 0$ at equlibrium

$$0 = \frac{q}{m}\left(E + V \times B\right) - \gamma V$$

now picking out components

$$E = E_0 \hat{x}, \quad B = B_0 \hat{z}, \quad V = V_x \hat{x} + V_y \hat{y}$$

$$0 = \frac{q}{m}\left(E_0 + V_y B_0\right) - \gamma V_x$$

$$0 = \frac{q}{m}\left(0 - V_x B_0\right) - \gamma V_y$$

or oops lets use $E = E_x \hat{x} + E_y \hat{y}$

this will mak it easy to handle hall voltage later

... so $E = E_x \hat{x} + E_y \hat{y}$, $B = B_0 \hat{z}$, $V = V_x \hat{x} + V_y \hat{y}$

$$0 = \frac{q}{m}\left(E_x + V_y B_0\right) - \gamma V_x$$

$$0 = \frac{q}{m}\left(E_y - V_x B_0\right) - \gamma V_y$$

$0 = 0$ trivially considering $\hat{z}$ force balance

$$\Downarrow \qquad -\frac{qB_0}{m} V_y + \gamma V_x = \frac{qE_x}{m}$$

$$\underbrace{\frac{qB_0}{m}}_{\Omega_c} V_x + \gamma V_y = \frac{qE_y}{m}$$

re-writing

$$\gamma V_x - \Omega_c V_y = \frac{qE_x}{m}$$

$$\Omega_c V_x + \gamma V_y = \frac{qE_y}{m}$$

$$\Downarrow$$

$$V_x = \left(\gamma E_x + \Omega_c E_y\right) \frac{(q/m)}{(\gamma^2 + \Omega_c^2)}$$

$$V_y = \left(-\Omega_c E_x + \gamma E_y\right) \frac{(q/m)}{(\gamma^2 + \Omega_c^2)}$$

or

$$\begin{pmatrix} V_x \\ V_y \end{pmatrix} = \frac{q/m}{(\gamma^2 + \Omega_c^2)} \begin{pmatrix} \gamma & \Omega_c \\ -\Omega_c & \gamma \end{pmatrix} \begin{pmatrix} E_x \\ E_y \end{pmatrix}$$

multiplying by nq, $\vec{j} = nq\vec{V}$

$$\vec{j} = \frac{nq^2}{m} \frac{1}{(\gamma^2 + \Omega_c^2)} \begin{pmatrix} \gamma & \Omega_c \\ -\Omega_c & \gamma \end{pmatrix} \vec{E}$$

$$\sigma_0 = \frac{q^2 n}{m\gamma} = \frac{q^2 n \tau}{m}$$

$$\vec{j} = \frac{\sigma_0}{1+\Omega_c^2\tau^2}\begin{pmatrix} 1 & \Omega\tau \\ -\Omega\tau & 1 \end{pmatrix} \vec{E}$$

or adding in the previously trivial $\hat{z}$ component

$$\vec{j} = \sigma_0 \begin{pmatrix} \frac{1}{1+\Omega^2\tau^2} & \frac{\Omega\tau}{1+\Omega^2\tau^2} & 0 \\ \frac{-\Omega\tau}{1+\Omega^2\tau^2} & \frac{1}{1+\Omega^2\tau^2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \vec{E}$$

now consider just an off diagonal component

$$\sigma_{xy} = \frac{\sigma_0 \Omega_c \tau}{1+\Omega_c^2\tau^2}$$

consider applying a strong B_z

ie $\Omega \gg \gamma = 1/\tau \Rightarrow \Omega\tau \gg 1$

$$\sigma_{xy} = \frac{\sigma_0 \Omega_c \tau}{1+\Omega_c^2\tau^2} = \frac{q^2 n \tau}{m} \frac{\Omega_c \tau}{1+\Omega_c^2\tau^2} \approx \frac{q^2 n}{m\Omega}$$

So σ_{xy} is usefull to explore material properties, but it is inconvenient to measure some off axis (DC E field) current.

So consider now a finite conductor. Charge must build up and creat some E_y such that $j \parallel \hat{x}$ steady state is achived. E_x is applied and E_y builds up so that carriers are confined to conductor

considering the 2D written

$$\vec{j} = \frac{\sigma_0}{1+\Omega^2\tau^2}\begin{pmatrix} 1 & \Omega\tau \\ -\Omega\tau & 1 \end{pmatrix}\vec{E}$$

$$\vec{j} = \begin{pmatrix} j_x \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} j_x \\ 0 \end{pmatrix} = \frac{\sigma_0}{1+\Omega^2\tau^2}\begin{pmatrix} 1 & \Omega\tau \\ -\Omega\tau & 1 \end{pmatrix}\begin{pmatrix} E_x \\ V_H/L_y \end{pmatrix}$$

$$V_{Hall} = \underbrace{(\text{Width of conductor})}_{L_y} E_y$$

the $j_y = 0$ row gives us

$$0 = -\Omega\tau E_x + V_H/L_y \Rightarrow V_H = \Omega\tau L E_x$$

putting this in the first row

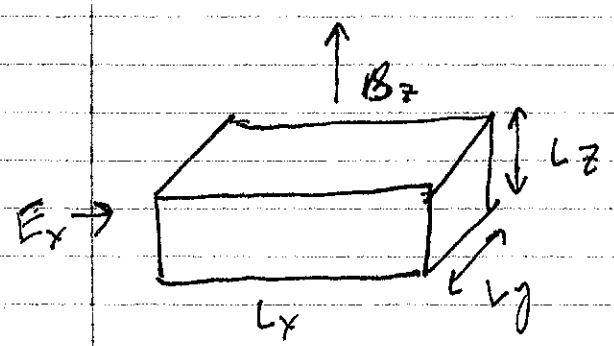
$$j_x = \frac{\sigma_0}{1+\Omega^2\tau^2}\left(E_x + \frac{\Omega^2\tau^2}{L} L_y E_x\right) = \sigma_0 E_x$$

so usual current does not depend on perpindicular magnetic field

but $V_H = \Omega\tau L_y E_x$ still gives us the sign of charge carrier

(sign of Ω depends on sign of q)

we can put this in macroscopic context and define



$$I_{tot} = j_x L_y L_z$$

and define $R_H = \dfrac{V_H}{I_{tot}}$

$$R_H = \frac{V_H}{I_{tot}} = \frac{\Omega\tau L_y E_x}{\sigma_0 E_x L_y L_z} = \frac{qB\tau}{m}\left(\frac{1}{q^2 n\tau/m}\right)\frac{L_y}{L_y L_z}\frac{E_x}{E_x} = \frac{B}{qnL_z}$$